

Ex. No. : 08

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A PYTHON PROGRAM TO IMPLEMENT BOOSTING

Aim:

To implement a **boosting algorithm (AdaBoost)** on the Iris dataset.

Requirements:

Jupyter Notebook with Python, Iris.csv dataset, and required Python libraries.

PROGRAM:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import sklearn

from sklearn.ensemble import AdaBoostClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import LabelEncoder

print("sklearn version:", sklearn.__version__)

from google.colab import files
uploaded = files.upload()

df = pd.read_csv('Iris.csv')

X = df[['SepalLengthCm', 'SepalWidthCm']]

le = LabelEncoder()
y = le.fit_transform(df['Species'])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
base_model = DecisionTreeClassifier(max_depth=1)
```

```
model = AdaBoostClassifier(  
    estimator=base_model,  
    n_estimators=50  
)
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)  
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
model.fit(X, y)
```

```
X_vis = X.values
```

```
x_min, x_max = X_vis[:,0].min()-1, X_vis[:,0].max()+1  
y_min, y_max = X_vis[:,1].min()-1, X_vis[:,1].max()+1
```

```
xx, yy = np.meshgrid(  
    np.linspace(x_min, x_max, 200),  
    np.linspace(y_min, y_max, 200)  
)
```

```
grid = np.c_[xx.ravel(), yy.ravel()]
```

```
Z = model.predict(grid)  
Z = Z.reshape(xx.shape)
```

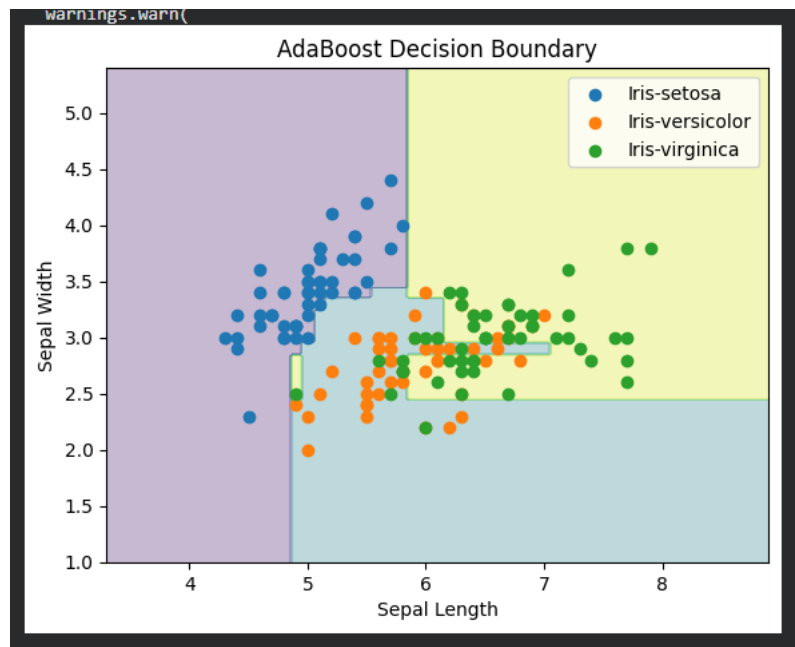
```
plt.figure()  
plt.contourf(xx, yy, Z, alpha=0.3)
```

```
for i, label in enumerate(np.unique(y)):  
    plt.scatter(  
        X_vis[y == label, 0],  
        X_vis[y == label, 1],  
        label=le.inverse_transform([label])[0]  
    )
```

```
plt.xlabel("Sepal Length")  
plt.ylabel("Sepal Width")  
plt.title("AdaBoost Decision Boundary")
```

```
plt.legend()  
plt.show()
```

OUTPUT:



Result:

Thus, the boosting model successfully improved classification performance and effectively identified fraudulent transactions.